

James Nithil V

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EXPERIENCE

Junior Software Developer

Aug 2025 to Present

Vaken Technologies Pvt Ltd | Product: Sovablu

Trichy, India

- Engineered and scaled enterprise RESTful APIs using **Java 21, Spring Boot, and PostgreSQL**, implementing strict **Spring Transaction management and ACID principles** across critical mutation and confirmation workflows to maintain data consistency and prevent data loss under concurrent client traffic.
- Architected a hybrid persistence layer pairing **PostgreSQL** with **Amazon DynamoDB** for fast key value reads. Restructured table designs and eliminated redundant columns to reduce I/O payload size while implementing **composite indexing and query optimization** to eliminate full row scans.
- Implemented an in memory **Redis caching tier with TTL expiration** for frequently queried datasets to reduce repeated database queries. Applied **cache stampede prevention mechanisms** using mutex locks to safeguard the database from concurrent traffic spikes and improve API response times.
- Secured file and document storage using **Amazon S3 with expiring presigned URLs** and enforced **role based access control (RBAC)** across platform endpoints. Automated Java compilation, testing, and containerized deployment to **Amazon ECS** via **Jenkins CI/CD** with **AWS Lambda** for event driven scheduled tasks.
- Contributed to **Agentic AI development** by building autonomous agents, defining modular skills, integrating **MCP tool access**, and deploying **RAG pipelines with guardrails**. Optimized LLM inference costs and response efficiency through **dynamic model routing**, directing routine operations to lightweight models and reserving complex reasoning for advanced LLMs.

PROJECTS

KadalVazhi: Maritime Operations and AI Platform

2026

Java 25, Spring Boot, Apache Kafka, Python, PostgreSQL, Redis, Offline SQLite Sync

Production Microservices

- Architected a distributed maritime operations and direct catch marketplace using **Java 25 and Spring Boot**, bypassing dockside middlemen through preorders to secure **15% to 25% higher profit margins** for deep sea fishing fleets.
- Designed a resilient **offline first synchronization engine with embedded SQLite** in Write Ahead Logging mode, enabling uninterrupted logging of boat telemetry, fuel, and catch data during **7 to 15 day deep sea cellular blackouts** with idempotent cloud reconciliation upon reaching coastal 4G.
- Built an event driven messaging pipeline with **Apache Kafka partitioned by Harbour ID** handling concurrent vessel telemetry, live dockside price broadcasts, and automated **emergency crew and marine mechanic dispatch** across coastal Tamil Nadu and Kerala.
- Integrated hands free **multilingual voice advisory in Tamil and Malayalam** powered by **Whisper Voice Engine and Gemini API**, cutting LLM inference costs by **45%** through intelligent dynamic model routing for routine marine navigation and weather queries.

TECHNICAL SKILLS

Languages: Java (Core & 21), Python, SQL, C++, JavaScript

Backend & APIs: Spring Boot, FastAPI, REST APIs, GraphQL, gRPC, Microservices

Distributed Systems & Cloud: Apache Kafka, Distributed Systems, Redis Caching, Rate Limiting, Docker, Jenkins CI/CD, AWS (EC2, S3, Lambda, Bedrock)

Databases: PostgreSQL, DynamoDB, MySQL, Redis, SQLite

AI & Systems: Agentic AI, MCP Tools, Claude SDK, Whisper Voice Engine, Gemini API, RAG, i18n

ACHIEVEMENTS

- LeetCode Guardian Badge:** Earned an official **Contest Rating of 2,134** (Top 1.33% worldwide) with **700+ algorithmic problems** solved in **Java and Python** (Graphs, Trees, Dynamic Programming, Sliding Window, Two Pointers, Hash Maps, Linked Lists).
- Smart India Hackathon (SIH) Team Lead:** Led a 5 member engineering team to build **FarmVista**, an AI precision agriculture platform integrating deep learning computer vision for automated crop disease diagnosis and soil telemetry.

EDUCATION

Knowledge Institute of Technology

Salem, India

B.Tech in Artificial Intelligence and Data Science | CGPA: 8.1

2021 to 2025

CERTIFICATIONS

Core Java & SQL (EBox) | MySQL Database (IBM) | Machine Learning (Simplilearn) | Deep Learning PyTorch (Guvi)